Q3 Constant acceleration equations

Name: ____________________

Class: ____________________

Date: ____________________

Time: **448 minutes**

Marks: **373 marks**

Comments:

Q1.

A sprinter accelerates from rest at a constant rate for the first 10 metres of a 100-metre race. He takes 2.5 seconds to run the first 10 metres.

(a) Find the acceleration of the sprinter during the first 2.5 seconds of the race.

(3)

(b) Show that the speed of the sprinter at the end of the first 2.5 seconds of the race is 8 m s^{-1}.

(2)

(c) The sprinter completes the 100-metre race, travelling the remaining 90 metres at a constant speed of 8 m s^{-1}. Find the total time taken for the sprinter to travel the 100 metres.

(3)

(d) Calculate the average speed of the sprinter during the 100-metre race.

(2)

(Total 10 marks)

Q2.

A ball is released from rest at a height of 15 metres above ground level.

(a) Find the speed of the ball when it hits the ground, assuming that no air resistance acts on the ball.

(3)

(b) In fact, air resistance does act on the ball. Assume that the air resistance force has a constant magnitude of 0.9 newtons. The ball has a mass of 0.5 kg.

(i) Draw a diagram to show the forces acting on the ball, including the magnitudes of the forces acting.

(1)

(ii) Show that the acceleration of the ball is 8 m s^{-2}.

(3)

(iii) Find the speed at which the ball hits the ground.

(2)

(iv) Explain why the assumption that the air resistance force is constant may not be valid.

(1)

(Total 10 marks)

Q3.

The constant forces $\mathbf{F}_1 = (8\mathbf{i} + 12\mathbf{j})$ newtons and $\mathbf{F}_2 = (4\mathbf{i} - 4\mathbf{j})$ newtons act on a particle. No other forces act on the particle.

(a) Find the resultant force acting on the particle.

(2)

(b) Given that the mass of the particle is 4 kg, show that the acceleration of the particle is $(3\mathbf{i} + 2\mathbf{j})$ m s^{-2}.

(2)

(c) At time t seconds, the velocity of the particle is $\mathbf{v}$ m s^{-1}.

(i) When $t = 20$, $\mathbf{v} = 40\mathbf{i} + 32\mathbf{j}$.

Show that $\mathbf{v} = -20\mathbf{i} - 8\mathbf{j}$ when $t = 0$.

(3)

(ii) Write down an expression for $\mathbf{v}$ at time t.

(1)

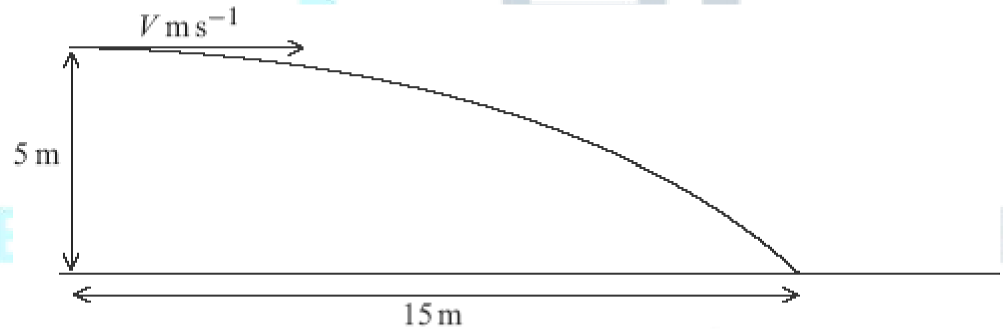
(iii) Find the times when the speed of the particle is 8 m s^{-1}.

(6)

(Total 14 marks)

Q4.

A ball is projected horizontally with speed V m s^{-1} at a height of 5 metres above horizontal ground. When the ball has travelled a horizontal distance of 15 metres, it hits the ground.



(a) Show that the time it takes for the ball to travel to the point where it hits the ground is 1.01 seconds, correct to three significant figures.

(3)

(b) Find V.

(2)

(c) Find the speed of the ball when it hits the ground.

(4)

(d) Find the angle between the velocity of the ball and the horizontal when the ball hits the ground. Give your answer to the nearest degree.

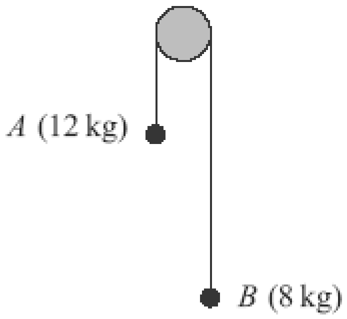
(3)

(e) State two assumptions that you have made about the ball while it is moving.

(2)
(Total 14 marks)

Q5.

Two particles, A and B, have masses 12 kg and 8 kg respectively. They are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram.



The particles are released from rest and move vertically. Assume that there is no air resistance.

(a) By forming two equations of motion, show that the magnitude of the acceleration of each particle is 1.96 m s^{-2}.

(5)

(b) Find the tension in the string.

(2)

(c) After the particles have been moving for 2 seconds, both particles are at a height of 4 metres above a horizontal surface. When the particles are in this position, the string breaks.

(i) Find the speed of particle A when the string breaks.

(2)

(ii) Find the speed of particle A when it hits the surface.

(3)
(Total 12 marks)

Q6.

A ship moves with constant acceleration. At time t seconds, its velocity is $\mathbf{v}$ m s^{-1}, where

$$\mathbf{v} = (9 - 0.01t)\mathbf{i} + (7 - 0.03t)\mathbf{j}$$

The unit vectors **i** and **j** are directed east and north respectively.

(a) Write down the velocity of the ship when $t = 0$.

(1)

(b) Find the acceleration of the ship.

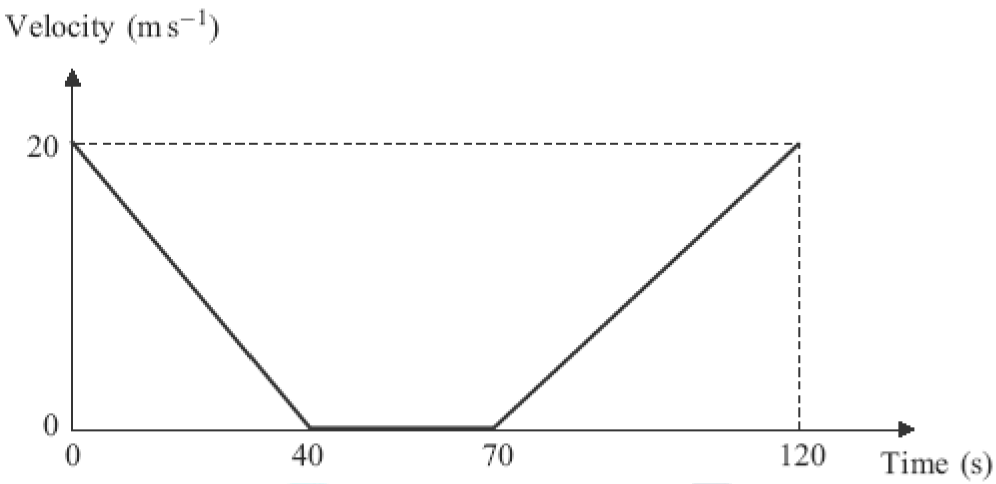
(2)

(c) Find t when the ship is travelling south-east.

(3)
(Total 6 marks)

Q7.

A bus slows down as it approaches a bus stop. It stops at the bus stop and remains at rest for a short time as the passengers get on. It then accelerates away from the bus stop. The graph shows how the velocity of the bus varies.



Assume that the bus travels in a straight line during the motion described by the graph.

(a) State the length of time for which the bus is at rest.

(1)

(b) Find the distance travelled by the bus in the first 40 seconds.

(2)

(c) Find the total distance travelled by the bus in the 120-second period.

(2)

(d) Find the average speed of the bus in the 120-second period.

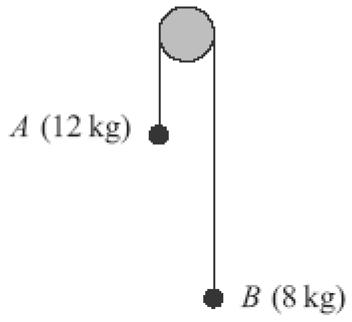
(2)

(e) If the bus had not stopped but had travelled at a constant 20 m s^{-1} for the 120-second period, how much further would it have travelled?

(2)
(Total 9 marks)

Q8.

Two particles, A and B, have masses 12 kg and 8 kg respectively. They are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram.



The particles are released from rest and move vertically. Assume that there is no air resistance.

(a) By forming two equations of motion, show that the magnitude of the acceleration of each particle is 1.96 m s^{-2}.

(5)

(b) Find the tension in the string.

(2)

(c) After the particles have been moving for 2 seconds, both particles are at a height of 4 metres above a horizontal surface. When the particles are in this position, the string breaks.

(i) Find the speed of particle A when the string breaks.

(2)

(ii) Find the speed of particle A when it hits the surface.

(3)

(iii) Find the time that it takes for particle B to reach the surface after the string breaks.
Assume that particle B does not hit the peg.

(5)

(Total 17 marks)

Q9.

A particle, of mass 10 kg, moves on a smooth horizontal surface. A single horizontal force, (9**i** + 12**j**) newtons, acts on the particle.

The unit vectors **i** and **j** are directed east and north respectively.

(a) Find the acceleration of the particle.

(2)

(b) At time t seconds, the velocity of the particle is **v** m s^{-1}. When t = 0, the velocity of the particle is (2.2**i** + **j**) m s^{-1} and the particle is at the origin.

(i) Find the distance between the particle and the origin when t = 5.

(4)

(ii) Express **v** in terms of t.

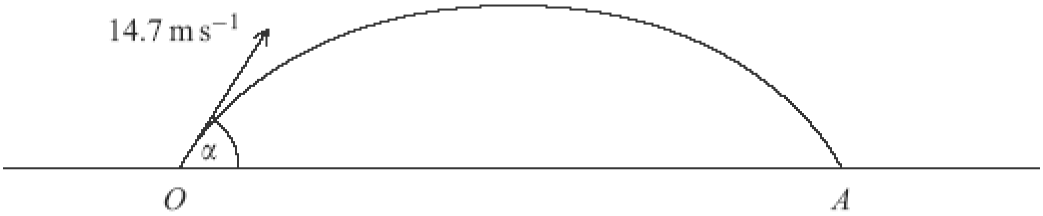
(2)

(iii) Find t when the particle is travelling north-east.

(3)
(Total 11 marks)

Q10.

A ball is struck so that it leaves a horizontal surface travelling at 14.7 m s^{-1} at an angle α above the horizontal. The path of the ball is shown in the diagram.



(a) Show that the ball takes $\dfrac{3 \sin \alpha}{2}$ seconds to reach its maximum height.

(3)

(b) The ball reaches a maximum height of 7 metres.

(i) Find α.

(5)

(ii) Find the range, OA.

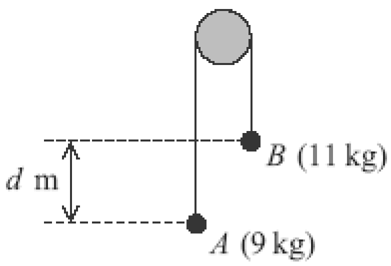
(3)

(c) State two assumptions that you needed to make in order to answer the earlier parts of this question.

(2)
(Total 13 marks)

Q11.

Two particles, A and B, are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram. The mass of A is 9 kg and the mass of B is 11 kg.



The particles are released from rest in the position shown, where B is d metres higher than A.
Assume that no resistance forces act on the particles.

(a) By forming an equation of motion for each of the particles A and B, show that the acceleration of each particle has magnitude 0.98 m s^{-2}.

(5)

(b) When the particles have been moving for 0.5 seconds, they are at the same level.

(i) Find the speed of the particles at this time.

(2)

(ii) Find d.

(4)
(Total 11 marks)

Q12.

A football is kicked from ground level with an initial velocity of 30 m s^{-1} at an angle of 35° above the horizontal.

(a) Find the maximum height of the ball above ground level.

(4)

(b) Show that when the speed of the ball is 28 m s^{-1}, the magnitude of the vertical component of its velocity is 13.4 m s^{-1}, correct to three significant figures.

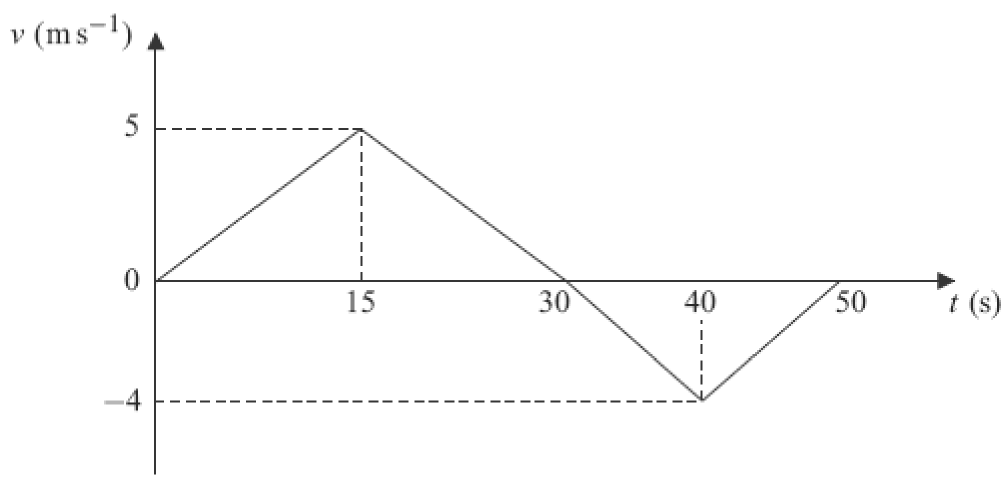
(4)

(c) Find the angle between the velocity of the ball and the horizontal when the speed of the ball is 28 m s^{-1}.

(2)
(Total 10 marks)

Q13.

The graph shows how the velocity of a particle varies during a 50-second period as it moves forwards and then backwards on a straight line.



(a) State the times at which the velocity of the particle is zero.

(2)

(b) Show that the particle travels a distance of 75 metres during the first 30 seconds of its motion.

(2)

(c) Find the total distance travelled by the particle during the 50 seconds.

(4)

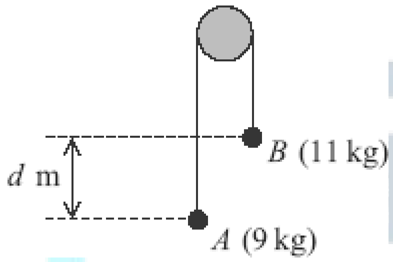
(d) Find the distance of the particle from its initial position at the end of the 50-second period.

(2)

(Total 10 marks)

Q14.

Two particles, A and B, are connected by a string that passes over a fixed peg, as shown in the diagram. The mass of A is 9 kg and the mass of B is 11 kg.



The particles are released from rest in the position shown, where B is d metres higher than A.
The motion of the particles is to be modelled using simple assumptions.

(a) State one assumption that should be made about the peg.

(1)

(b) State two assumptions that should be made about the string.

(2)

(c) By forming an equation of motion for each of the particles A and B, show that the acceleration of each particle has magnitude 0.98 m s^{-2}.

(5)

(d) When the particles have been moving for 0.5 seconds, they are at the same level.

(i) Find the speed of the particles at this time.

(2)

(ii) Find d.

(4)

(Total 14 marks)

Q15.

Two forces, **P** = (6**i** − 3**j**) newtons and **Q** = (3**i** + 15**j**) newtons, act on a particle. The unit vectors **i** and **j** are perpendicular.

(a) Find the resultant of **P** and **Q**.

(2)

(b) Calculate the magnitude of the resultant of **P** and **Q**.

(2)

(c) When these two forces act on the particle, it has an acceleration of (1.5**i** + 2**j**) m s^{-2}. Find the mass of the particle.

(2)

(d) The particle was initially at rest at the origin.

(i) Find an expression for the position vector of the particle when the forces have been applied to the particle for t seconds.

(2)

(ii) Find the distance of the particle from the origin when the forces have been applied to the particle for 2 seconds.

(2)

(Total 10 marks)

Q16.

A motorcycle and rider, of total mass 300 kg, are accelerating in a straight line along a horizontal road at 2.2 m s^{-2}.

(a) Show that the magnitude of the resultant force acting on the motorcycle is 660 N.

(1)

(b) A forward driving force of P newtons together with a resistance force of magnitude 400 newtons act on the motorcycle. Find P.

(2)

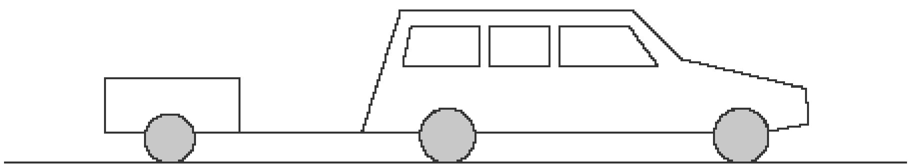
(c) Find the time that it would take for the speed of the motorcycle to increase from 12 m s^{-1} to 23 m s^{-1}.

(3)

(Total 6 marks)

Q17.

A car, of mass 1400 kg, is towing a trailer, of mass 600 kg. The two vehicles accelerate together at 1.3 m s^{-2} along a straight horizontal road.



(a) Find the distance that the car and trailer would travel while accelerating from rest to 13 m s^{-1}.

(3)

(b) A forward driving force, of magnitude 3900 N, acts on the car. A resistance force, of magnitude 800 N, also acts on the car.

(i) A resistance force, of magnitude P newtons, acts on the trailer. Find P.

(3)

(ii) Find the magnitude of the force that the car exerts on the trailer.

(3)

(Total 9 marks)

Q18.

A lift is travelling upwards and accelerating uniformly. During a 5 second period, it travels 16 metres and the speed of the lift increases from u m s^{-1} to 4.2 m s^{-1}.

(a) Find u.

(3)

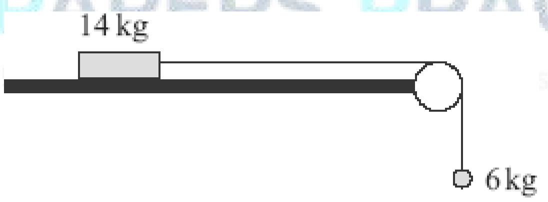
(b) Find the acceleration of the lift.

(3)

(Total 6 marks)

Q19.

A block, of mass 14 kg, is held at rest on a rough horizontal surface. The coefficient of friction between the block and the surface is 0.25. A light inextensible string, which passes over a fixed smooth peg, is attached to the block. The other end of the string is attached to a particle, of mass 6 kg, which is hanging at rest.



The block is released and begins to accelerate.

(a) Find the magnitude of the friction force acting on the block.

(3)

(b) By forming two equations of motion, one for the block and one for the particle, show that the magnitude of the acceleration of the block and the particle is 1.225 m s^{-2}.

(5)

(c) Find the tension in the string.

(2)

(d) When the block is released, it is 0.8 metres from the peg. Find the speed of the block when it hits the peg.

(3)

(e) When the block reaches the peg, the string breaks and the particle falls a further 0.5 metres to the ground. Find the speed of the particle when it hits the ground.

(3)

(Total 16 marks)

Q20.

A particle moves on a smooth horizontal plane. It is initially at the point A, with position vector
($9\mathbf{i} + 7\mathbf{j}$) m, and has velocity ($-2\mathbf{i} + 2\mathbf{j}$) m s^{-1}. The particle moves with a constant acceleration of ($0.25\mathbf{i} + 0.3\mathbf{j}$) m s^{-2} for 20 seconds until it reaches the point B. The unit vectors **i** and **j** are directed east and north respectively.

(a) Find the velocity of the particle at the point B.

(3)

(b) Find the velocity of the particle when it is travelling due north.

(4)

(c) Find the position vector of the point B.

(3)

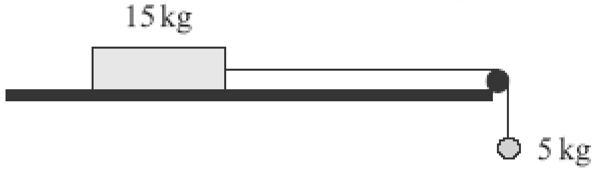
(d) Find the average velocity of the particle as it moves from A to B.

(2)

(Total 12 marks)

Q21.

A block, of mass 15 kg, is placed on a rough horizontal surface. It is attached, by a light inextensible string that passes over a smooth peg, to a particle of mass 5 kg, which hangs freely, as shown in the diagram.



The coefficient of friction between the block and the surface is 0.2. The block and particle are released from rest and move with the string taut.

(a) Find the magnitude of the friction force acting on the block.

(3)

(b) Show that the acceleration of the system is 0.98 m s^{-2}.

(5)

(c) The block is travelling at 1.4 m s^{-1} when it reaches the peg.

Find the distance that the block has travelled when it reaches the peg.

(3)
(Total 11 marks)

Q22.

A crane is used to lift a crate, of mass 70 kg, vertically upwards. As the crate is lifted, it accelerates uniformly from rest, rising 8 metres in 5 seconds.

(a) Show that the acceleration of the crate is 0.64 m s^{-2}.

(2)

(b) The crate is attached to the crane by a single cable. Assume that there is no resistance to the motion of the crate.

Find the tension in the cable.

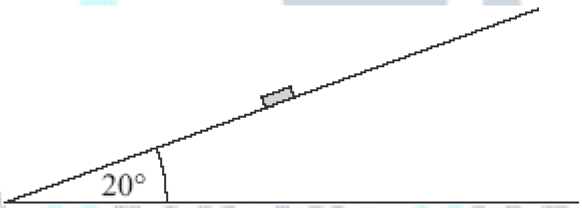
(3)

(c) Calculate the average speed of the crate during these 5 seconds.

(1)
(Total 6 marks)

Q23.

A puck, of mass 0.2 kg, is placed on a slope inclined at 20° above the horizontal, as shown in the diagram.



The puck is hit so that initially it moves with a velocity of 4 m s^{-1} directly up the slope.

(a) A simple model assumes that the surface of the slope is smooth.

(i) Show that the acceleration of the puck up the slope is -3.35 m s^{-2}, correct to three significant figures.

(3)

(ii) Find the distance that the puck will travel before it comes to rest.

(3)

(iii) What will happen to the puck after it comes to rest?

Explain why.

(2)

(b) A revised model assumes that the surface is rough and that the coefficient of friction between the puck and the surface is 0.5.

(i) Show that the magnitude of the friction force acting on the puck during this motion is 0.921 N, correct to three significant figures.

(3)

(ii) Find the acceleration of the puck up the slope.

(3)

(iii) What will happen to the puck after it comes to rest in this case?

Explain why.

(2)

(Total 16 marks)

Q24.

A Jet Ski is at the origin and is travelling due north at 5 m s^{-1} when it begins to accelerate uniformly. After accelerating for 40 seconds, it is travelling due east at m s^{-1}. The unit vectors **i** and **j** are directed east and north respectively.

(a) Show that the acceleration of the Jet Ski is (0.1 **i** – 0.125 **j**) m s^{-2}.

(4)

(b) Find the position vector of the Jet Ski at the end of the 40 second period.

(3)

(c) The Jet Ski is travelling southeast t seconds after it leaves the origin.

(i) Find t.

(5)

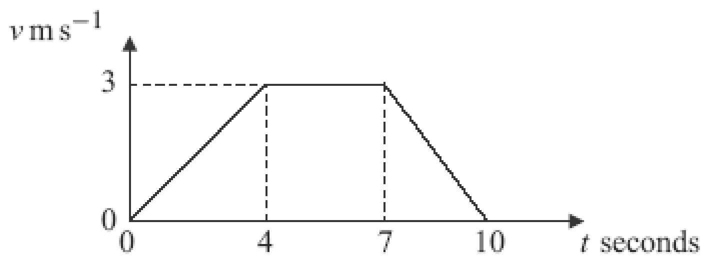
(ii) Find the velocity of the Jet Ski at this time.

(2)

(Total 14 marks)

Q25.

The diagram shows a velocity–time graph for a lift.



(a) Find the distance travelled by the lift.

(3)

(b) Find the acceleration of the lift during the first 4 seconds of the motion.

(1)

(c) The lift is raised by a single vertical cable. The mass of the lift is 400 kg. Find the tension in the cable during the first 4 seconds of the motion.

(3)
(Total 7 marks)

Q26.

The unit vectors **i** and **j** are directed east and north respectively. A helicopter moves horizontally with a constant acceleration of $(-0.4\mathbf{i} + 0.5\mathbf{j})$ m s^{-2}. At time $t = 0$, the helicopter is at the origin and has velocity $20\mathbf{i}$ m s^{-1}.

(a) Write down an expression for the velocity of the helicopter at time t seconds.

(2)

(b) Find the time when the helicopter is travelling due north.

(3)

(c) Find an expression for the position vector of the helicopter at time t seconds.

(2)

(d) When $t = 100$:

(i) show that the helicopter is due north of the origin;

(3)

(ii) find the speed of the helicopter.

(3)
(Total 13 marks)

Q27.

A motorcycle accelerates uniformly along a straight horizontal road so that, when it has travelled 20 metres, its velocity has increased from 12 m s^{-1} to 16 m s^{-1}.

(a) Find the acceleration of the motorcycle.

(3)

(b) Find the time that it takes for the motorcycle to travel this distance.

(3)
(Total 6 marks)

Q28.

A lift rises vertically from rest with a constant acceleration.

After 4 seconds, it is moving upwards with a velocity of 2 m s^{-1}.

It then moves with a constant velocity for 5 seconds.

The lift then slows down uniformly, coming to rest after it has been moving for a total of 12 seconds.

(a) Sketch a velocity–time graph for the motion of the lift.

(4)

(b) Calculate the total distance travelled by the lift.

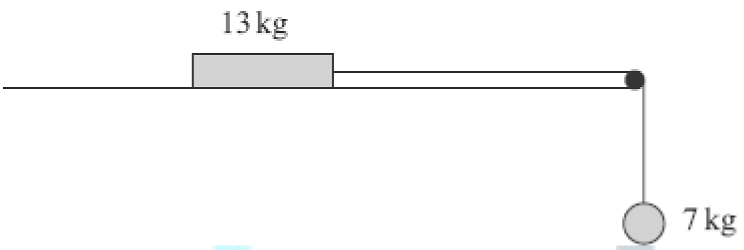
(2)

(c) The lift is raised by a single vertical cable. The mass of the lift is 300 kg. Find the maximum tension in the cable during this motion.

(4)
(Total 10 marks)

Q29.

The diagram shows a block, of mass 13 kg, on a rough horizontal surface. It is attached by a string that passes over a smooth peg to a sphere of mass 7 kg, as shown in the diagram.



The system is released from rest, and after 4 seconds the block and the sphere both have speed 6 m s^{-1}, and the block has **not** reached the peg.

(a) State **two** assumptions that you should make about the string in order to model the motion of the sphere and the block.

(2)

(b) Show that the acceleration of the sphere is 1.5 m s^{-2}.

(2)

(c) Find the tension in the string.

(3)

(d) Find the coefficient of friction between the block and the surface.

(6)
(Total 13 marks)

Q30.

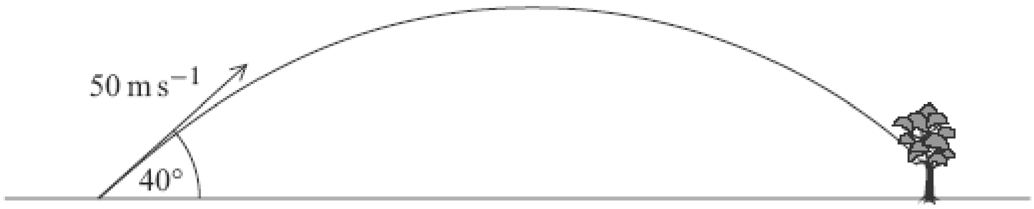
A golf ball is struck from a point on horizontal ground so that it has an initial velocity of 50 m s^{-1} at an angle of 40° above the horizontal.

Assume that the golf ball is a particle and its weight is the only force that acts on it once it is moving.

(a) Find the maximum height of the golf ball.

(4)

(b) After it has reached its maximum height, the golf ball descends but hits a tree at a point which is at a height of 6 metres above ground level.



Find the time that it takes for the ball to travel from the point where it was struck to the tree.

(6)

(Total 10 marks)

Q31.

A particle is initially at the origin, where it has velocity (5**i** – 2**j**) m s^{-1}. It moves with a constant acceleration **a** m s^{-2} for 10 seconds to the point with position vector 75**i** metres.

(a) Show that **a** = 0.5**i** + 0.4**j**.

(3)

(b) Find the position vector of the particle 8 seconds after it has left the origin.

(3)

(c) Find the position vector of the particle when it is travelling parallel to the unit vector **i**.

(6)

(Total 12 marks)

Q32.

A hot air balloon is at rest on the ground. When the balloon is released, it rises to a height of 320 metres in 80 seconds. The balloon moves under the action of its weight and a vertical lift force. Assume that the balloon has a constant acceleration during this motion.

(a) Show that the acceleration of the balloon is 0.1 m s^{-2}.

(3)

(b) Find the speed of the balloon when it reaches a height of 320 metres.

(2)

(c) The mass of the balloon is 450 kg. Show that the magnitude of the vertical lift force is 4500 N, correct to two significant figures.

(3)

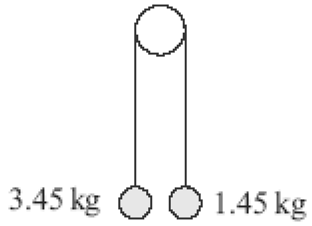
(d) After a while, the vertical lift force is reduced so that the balloon rises at a constant speed. State the magnitude of the vertical lift force when this is the case.

(1)

(Total 9 marks)

Q33.

Two particles, of masses 3.45 kg and 1.45 kg, are connected by a light string that passes over a smooth peg. The particles are released from rest with the strings vertical, as shown in the diagram.



(a) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 4 m s^{-2}.

(5)

(b) Find the tension in the string.

(2)

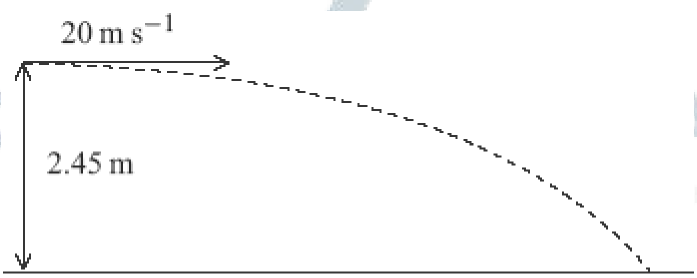
(c) Initially the particles are at the same level.

Find the speed of the heavier particle when it is 1 metre lower than the lighter particle. Assume that neither particle hits the floor or the peg.

(3)
(Total 10 marks)

Q34.

A tennis ball is hit from a height of 2.45 metres above horizontal ground. Initially it travels horizontally at a speed of 20 m s^{-1}, as shown in the diagram.



(a) Show that the time taken for the tennis ball to reach the ground is 0.707 seconds, correct to three significant figures.

(3)

(b) Find the horizontal distance travelled by the ball when it hits the ground.

(2)

(c) Find the angle between the velocity of the ball and the horizontal when the ball hits the ground.

(4)
(Total 9 marks)

Q35.

A ball is released from rest at a height h metres above ground level. The ball hits the ground 1.5 seconds after it is released. Assume that the ball is a particle that does not experience any air resistance.

(a) Show that the speed of the ball is 14.7 m s^{-1} when it hits the ground.

(2)

(b) Find h.

(2)

(c) Find the distance that the ball has fallen when its speed is 5 m s^{-1}.

(3)

(Total 7 marks)